NUMERICAL METHODS
CPE 312

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Lecturer

SECANT METHOD

Learning Objectives

*Define Secant Method

*Determine and perform the algorithm of Secant Method

*Solve functions using Secant Method

- The Secant Method is a root-finding procedure in numerical analysis that uses a series of roots of secant lines to better approximate a root of a function f.
- It's similar to the **Regular-Falsi** method but we don't need to check $\mathbf{f(x_1)f(x_2)<0}$ again and again after every approximation.

- In this method, the neighbourhoods roots are approximated by secant line or chord to the function **f(x)**.
- It's also advantageous of this method that we don't need to differentiate the given function **f(x)**, as we do in **Newton-raphson** method.

The method is particularly useful when:

1. Derivatives of the function are difficult to compute or don't exist.
2. Faster convergence than basic methods like bisection is desired.

Secant Method Algorithm

1. Find points x_0 and x_1 such that

$$x_0 < x_1 .$$

2. Take the interval $[x_0 , x_1]$ and find next value using

$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$

3. Repeat steps 2 until $f(x_2) \approx 0$.
Stopping criterion (0.000…)

Example-1

Approximate the root of $f(x) = x^3 + 2x^2 + x - 1$ using Secant Method.

$$f(x) = x^3 + 2x^2 + x - 1$$

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$$f(0) = 0^3 + 2(0)^2 + 0 - 1 = \mathbf{-1}$$

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Initial interval [0, 1]

	X_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	0		1			

	X_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	0	-1	1	3		

	X_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	0	-1	1	3	0.25	

	X_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	0	-1	1	3	0.25	-0.60938

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2	1	3	0.25	-0.60938	0.37662	

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5	0.48880	0.08344	0.46348	-0.00733	0.46552	-0.00018

The approximate root of
$f(x) = x^3 + 2x^2 + x - 1$ is 0.46552.

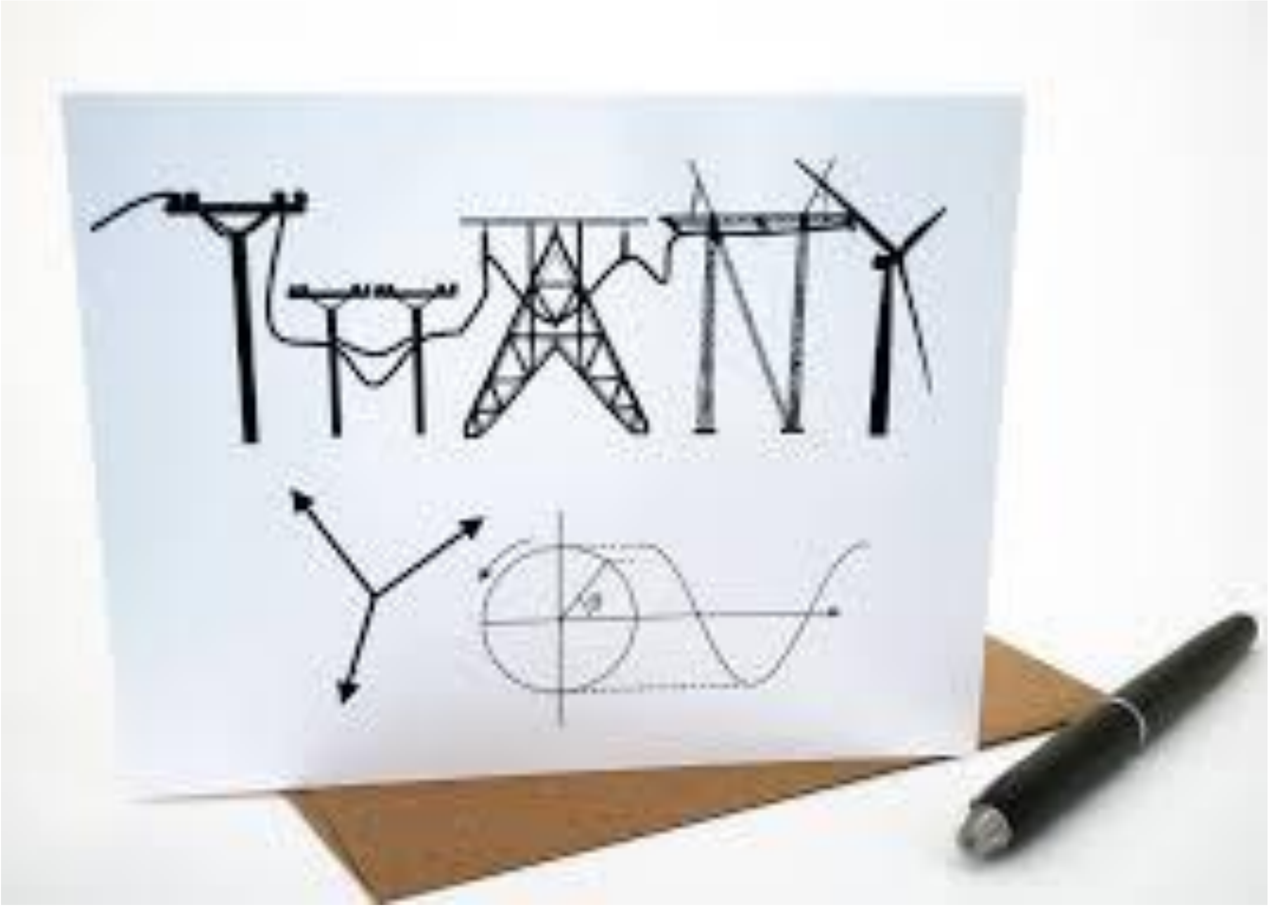
ACTIVITY # 7

1. Find the approximate root of $f(x) = \sin(x) - 0.5$ using Secant Method.

2. Use the Newton-Raphson Method to find the root of $f(x) = 2x^3 - 2x - 5$.

Reminder:

Submit your Activity # 7
https://forms.gle/Lvennc1HSmj97ysQ7
on or before 7:00 pm.



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